

Condensate — Where the Water Goes

The evaporator makes water every hour it runs. Most cooling defects you will ever write up are about that water.

Warm indoor air passing over a cold evaporator coil condenses on it, exactly the way a cold glass sweats. That water has to be collected and taken somewhere, continuously, for as long as the system runs.

Primary drain

Runs from the evaporator pan, with a trap. The trap stops the blower pulling air back up the drain line.

Auxiliary drain pan

The second pan under the unit, for when the primary fails. Asked which statements about auxiliary drain pans are true, the answer is all of them — they need their own drain, that drain must discharge somewhere visible, and the pan must be sloped and leak-free. The options are separate requirements, not competing candidates.

YOU HAVE MISSED THIS

Which of the following is TRUE concerning auxiliary drain pans?

All of the answers are true.

The whole point of the auxiliary pan is to announce a failure somewhere you will see it, before the ceiling does the announcing.

Reading a ceiling stain

Part of the air conditioning unit is in the attic and there is moisture on the ceiling below. The likely cause is the evaporator coil drain pan. Work backwards from where the water is: an attic air handler above a wet ceiling means the pan is full, cracked or its drain is blocked.

The best way to drain condensation from the evaporator coil is a plastic line running to a floor drain, to the exterior, or to the sink side of a trap. The sink side matters — discharging into the sewer side of a trap would let sewer gas back into the air handler.

Say it back

Cover the page. Answer each one out loud.

1. Why does an evaporator coil produce water?
2. What does the trap in the primary drain line stop?
3. What is true about auxiliary drain pans, and why is the answer cumulative?
4. Attic air handler, wet ceiling below. Most likely cause?
5. Name the three acceptable destinations for a condensate line.
6. Why the sink side of the trap and not the other side?